UGV PROJECT

IIT HYDERABAD

EE1010

MATHS FOR ELECTRICAL ENGINEERING

Prof. G. V. V. SHARMA

SK. M M SUHAN — **EE26BTECH11228**

Y. VISWA SRIHAS — **EE26BTECH11234**

BLOCK DIAGRAM:

Dabble app → Bluetooth → ESP32 → L293D → DC motors → Car movement

1.OBJECTIVE:

To develop a Bluetooth-controlled Unmanned Ground Vehicle (UGV) using an ESP32 microcontroller, an L293D motor driver, and DC motors, allowing the user to wirelessly control the vehicle's forward, backward, left, and right movements through the Dabble Gamepad mobile application.

2.COMPONENTS AND THEIR FUNCTIONS:

COMPONENTS	FUNCTIONS
ESP32 Microcontroller	Receives movement commands from the Dabble mobile application through Bluetooth and controls the motors by sending signals to the L293D motor driver.
L293D Motor Driver IC	Controls the direction of rotation of the DC motors according to the signals received from the ESP32. It allows the ESP32 to control the motors without driving them directly.
DC MOTORS	Convert electrical energy into mechanical rotation to drive the vehicle's wheels and produce movement.
DABBLE MOBILE APPLICATION	Provides a virtual Gamepad interface to send forward, backward, left, and right movement commands from the mobile phone.
CHASSIS	Acts as the main frame of the vehicle and supports the motors, wheels, ESP32, motor driver, and other components.
WHEELS	Convert the rotational motion of the DC motors into the movement of the vehicle.
Breadboard	Allows temporary, solderless connections between the ESP32, L293D IC, and jumper wires during circuit assembly.
JUMPER WIRES	Establish electrical power to the ESP32 , L293D motor driver , power supply , and other circuit components.

POWER BANK	Supplies electrical power to the ESP32 through its USB connection. The motors also require a suitable power supply that meets their voltage and current requirements.
USB Cable	Connects the ESP32 to the power bank for power or to a computer for programming and debugging

3.PROGRESS LOG OF UGV PROJECT:

1. **First, we took a breadboard and placed L293D motor driver IC on it.**
2. **We tried to connect esp32 to the ic by using female jumper wires but that made the positioning of esp32 on board complex.**
3. **So we planned to keep esp32 on the breadboard and used male jumper wires.**
4. **Next, we connected the DC motor wires to the output pins of the L293D motor driver IC. The L293D controls the direction of rotation of the motors based on the signals received from the ESP32, enabling the UGV to move in different directions.**
5. **Next, we connected the ESP32 microcontroller to the L293D motor driver IC using jumper wires. These connections allow the ESP32 to send control signals to the L293D, enabling it to control the direction of rotation of the DC motors.**

6. CONNECTIONS OF ESP32 AND L293D

ESP32	L293D
GPIO32	PIN 2
GPIO 33	PIN 7
GPIO 25	PIN 10
GPIO 26	PIN 15
GND	GND (PIN 4,5,12,13)

7. L293D TO DC MOTOR CONNECTIONS

L293D PIN	CONNECTION
PIN3	1ST TERMINAL OF MOTOR 1(+ve)
PIN 6	2ND TERMINAL OF MOTOR 1(-ve)
PIN 11	1ST TERMINAL OF MOTOR 2(-ve)
PIN 14	2ND TERMINAL OF MOTOR 2(+ve)

8. We uploaded a test code by connecting the esp32 to laptop.
9. We uploaded a test code that makes the wheels rotate forward for a certain duration, stop, then rotate backward for a certain duration, stop, and repeat the same cycle continuously.
10. We realised that one wheel is not rotating properly.
11. So, we soldered the wires of motor again and started to work.
12. We uploaded the code for UGV working to esp32 by holding boot button while uploading and connected that to mobile via Bluetooth using dabble app.
13. Right button started to work as front, front button as left, and so on.
14. We changed the code and uploaded the new code.
15. UGV started to work properly after that.

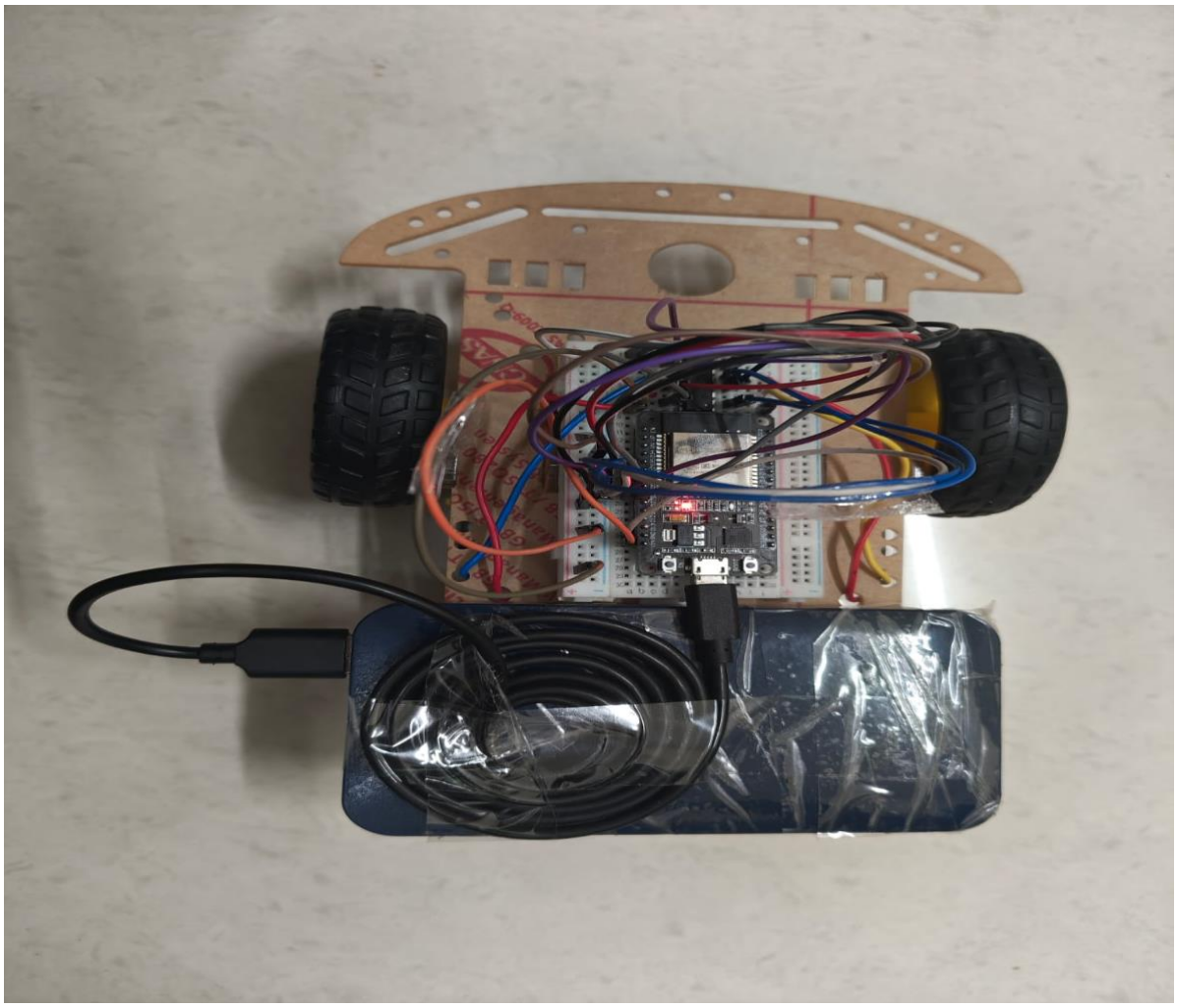
16. We enhanced the UGV navigation by adding joystick controls in addition to the D-pad by making modifications in our code.

17. The UGV is powered by a portable power bank.

18. We've finalized our Bluetooth-connected UGV, fully controllable using the Dabble app's joystick and D-pad interfaces.







4.References

a) prof. G. V. V Sharma (main pdf)

b) Chatgpt, Gemini

Chatgpt – for connections and understanding

Gemini and chatgpt – for codes